

10-minute conference oral

Attention Is All You Need

Sequence transduction without recurrence

Paper by Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit,
Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin

NeurIPS 2017 · [arXiv:1706.03762v7](https://arxiv.org/abs/1706.03762v7)

Presented by Anonymous Presenter

Must sequence modeling be sequential?

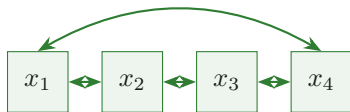
Self-attention lets positions exchange information directly within a layer.

Recurrence: follow the chain



Each state waits for the previous state.
Distant signals traverse many steps.

Self-attention: connect positions



Compute token interactions together.
Distant positions connect directly.

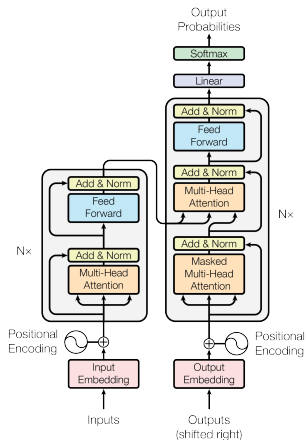
The paper's question

Can attention replace recurrence and convolution for machine translation?

Vaswani et al., Sections 1 and 4. Illustrations show selected connections, not full architectures.

The Transformer: an attention-based architecture

The complete original architecture; the next pages unpack its attention blocks.



Read from the bottom up

1. Embeddings + position information.
2. Encoder: source tokens interact.
3. Decoder: read the prefix and encoder outputs.
4. Predict the next token.

Base model: 6 layers in each stack;

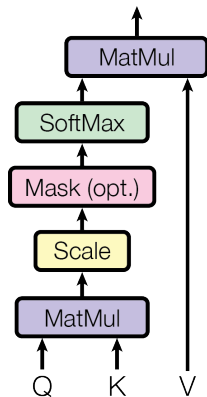
$$d_{\text{model}} = 512.$$

Add & Norm: residual addition, then layer normalization.

Original Figure 1, Vaswani et al. (2017), v7.
Colors, arrows, and notation unchanged.

Attention retrieves a weighted mixture of values

Queries match keys; the resulting weights combine values.



Original Figure 2, left panel.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

Q – queries What each position is looking for.

K – keys What each position can be matched on.

V – values The information to retrieve.

Divide by $\sqrt{d_k}$ to control dot-product scale.

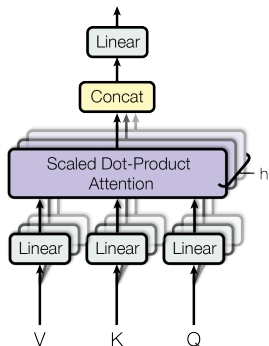
Softmax turns scores into weights over key positions.

Vaswani et al., Section 3.2.1, Eq. (1).

The optional mask is applied before softmax.

Multiple heads learn different projections

Each head has its own attention weights; their outputs are combined.



1. Project Q , K , and V for each head.
2. Run attention heads in parallel.
3. Concatenate, then apply an output projection.

Base model

$h = 8$ heads $d_k = d_v = 64$

$8 \times 64 = 512$ output features before the final projection.

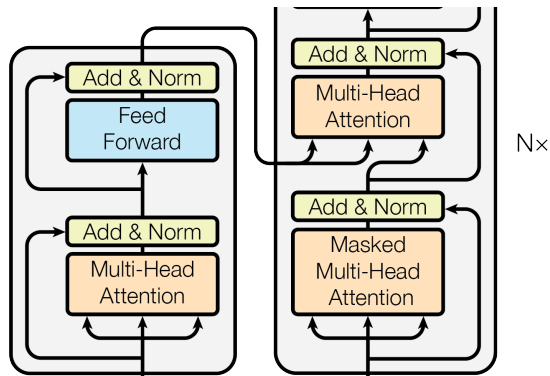
Vaswani et al., Section 3.2.2.

Head roles are learned, not assigned in advance.

Original Figure 2, right panel.

The decoder reads both the prefix and the source

The middle decoder attention block gets keys and values from the final encoder outputs.



Original Figure 1, enlarged middle/lower stacks.
Top decoder layers and embeddings lie outside this crop.

Bottom right: masked self-attention

Q, K, V : decoder representations.
Access only the available prefix.

Middle right: cross-attention

Q : decoder state.
 K, V : final encoder outputs.
Attend across all source positions.
Vaswani et al., Sections 3.1 and 3.2.3.

Position and masking solve different problems

Positions encode order; the causal mask prevents access to future targets.

Where is this token?

$$\text{input}_i = \text{embedding}_i + PE_i$$

The paper uses sine/cosine positional encodings in both stacks.

What may this token read?

Shift target inputs right and mask future positions before softmax.

Decoder inputs (shifted right)

	BOS	y_1	y_2	y_3
predict y_1	✓	×	×	×
predict y_2	✓	✓	×	×
predict y_3	✓	✓	✓	×
predict y_4	✓	✓	✓	✓

✓ allowed × blocked BOS: start token

Training: all target positions together. **Generation:** still one token at a time.

Vaswani et al., Sections 3.1 and 3.5. Mask grid: supplemental presenter illustration.

Test translation quality and training cost

WMT 2014 translation; historical comparisons with reported training budgets.

Data and evaluation

English–German: ~ 4.5 M sentence pairs.

English–French: 36M sentence pairs.

Test set: **newstest2014**.

BLEU $\uparrow$; beam size 4.

Checkpoint averaging for test results.

One machine, 8 NVIDIA P100 GPUs

Base: 100K steps, about 12 hours.

Big: 300K steps, about 3.5 days.

Training FLOPs are estimates from time, GPU count, and GPU throughput.

Different systems used different budgets.

Read Table 2 as a quality/cost comparison across published systems.

Vaswani et al., Sections 5.1–5.2 and 6.1. Main result discussion focuses on English–German.

28.4 BLEU on English–German

Transformer big exceeds the strongest prior EN–DE row by 2.04 BLEU points.

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	41.29	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	$3.3 \cdot 10^{18}$	
Transformer (big)	28.4	41.8	$2.3 \cdot 10^{19}$	

EN-DE BLEU

28.4

Transformer big

26.36

ConvS2S ensemble

+2.04 points

Base: 27.3 BLEU at an estimated 3.3×10^{18} training FLOPs.

Original Table 2, Vaswani et al., v7; newstest2014. Bracketed IDs belong to the paper.
EN–FR discrepancy: Table 2 / abstract report 41.8; Section 6.1 prose reports 41.0.

More heads are not always better

On the development set, 8 and 16 heads tie; 32 heads score lower.

	N	d_{model}	d_{ff}	h	d_k	d_v	P_{drop}	ϵ_{ls}	train steps	PPL (dev)	BLEU (dev)	params $\times 10^6$
base	6	512	2048	8	64	64	0.1	0.1	100K	4.92	25.8	65
(A)				1	512	512				5.29	24.9	
				4	128	128				5.00	25.5	
				16	32	32				4.91	25.8	
				32	16	16				5.01	25.4	

One head $\rightarrow$ eight heads

24.9 $\rightarrow$ 25.8: **+0.9 BLEU points.**

Head dimensions vary to keep computation comparable.

newstest2013 (dev); no checkpoint averaging.

Original Table 3, base row and group (A), Vaswani et al., v7.

Blank cells inherit the base settings. These scores are not the test scores on the previous page.

Eight / sixteen / thirty-two heads

25.8 / 25.8 / 25.4: a tie, then a decline.

Shorter paths, quadratic attention cost

The efficiency advantage depends on sequence length and the operation being measured.

Layer type	Interaction cost	Sequential steps	Maximum path
Self-attention	$O(n^2d)$	$O(1)$	$O(1)$
Recurrent	$O(nd^2)$	$O(n)$	$O(n)$

n : sequence length; d : representation width.

- Full attention compares all pairs of positions.
- These are layer-level comparisons, not end-to-end latency guarantees.
- Autoregressive decoding retains a token-by-token dependency.

Selected rows transcribed from Vaswani et al., Table 1; Sections 3.1, 4, and 7.

Full Transformer blocks also contain projections and feed-forward networks.

Attention can carry sequence transduction

The contribution is a complete architecture, supported by translation evidence.

1 Change how tokens communicate

Replace sequence-aligned recurrence with direct, learned attention.

2 Make the architecture usable

Multi-head attention, positions, masking, feed-forward layers, and residual paths.

3 Test quality and cost together

28.4 EN-DE BLEU; strong reported training efficiency, with explicit scope limits.

Thank you.

Vaswani et al., *Attention Is All You Need*, NeurIPS 2017.

Questions

Backup: the base model in precise terms

The 2017 architecture uses normalization after residual addition.

Dimensions

$N = 6$ per stack; $d_{\text{model}} = 512$.

$h = 8$; $d_k = d_v = 64$; $d_{\text{ff}} = 2048$.

Residual sublayer

$\text{LayerNorm}(x + \text{Sublayer}(x))$

Dropout on the sublayer output before addition.

Position-wise feed-forward network

$\max(0, xW_1 + b_1)W_2 + b_2$

Vaswani et al., Sections 3.1, 3.3, 3.5, 5.3, and 5.4.

Sinusoidal position encoding

$$PE_{(p,2i)} = \sin(p/10000^{2i/d_{\text{model}}})$$

$$PE_{(p,2i+1)} = \cos(p/10000^{2i/d_{\text{model}}})$$

p : token position; i : dimension-pair index.

Training

Adam; 4,000-step learning-rate warmup.

Base dropout: 0.1; label smoothing: 0.1.

Backup: keep the score conditions attached

The v7 English–French discrepancy remains unresolved in the paper.

Transformer big, English–French

Table 2 and abstract: **41.8 BLEU**. Section 6.1 prose: **41.0 BLEU**.

On the WMT 2014 English-to-French translation task, our big model achieves a BLEU score of 41.0, outperforming all of the previously published single models, at less than 1/4 the training cost of the previous state-of-the-art model. The Transformer (big) model trained for English-to-French used dropout rate $P_{drop} = 0.1$, instead of 0.3.

Test $\neq$ development

Table 2: newstest2014, checkpoint averaging.

Table 3: newstest2013, no checkpoint averaging.

Original prose excerpt: Vaswani et al., v7, Section 6.1, PDF p. 8.

Table 3 base / row (E); Section 6.2. No silent correction or averaging of the conflicting values.

Positional encoding ablation

Sinusoidal: 25.8 dev BLEU.

Learned: 25.7 dev BLEU.

Close reported scores; no significance claim.

Backup: reference and source version

All original figures and table excerpts come from the same paper version.

 Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin.
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Version consulted: [arXiv:1706.03762v7](https://arxiv.org/abs/1706.03762v7).

<https://arxiv.org/abs/1706.03762v7>

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